

AWS Cloud Architecture & Security Review

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Purpose

This portfolio sample demonstrates structured technical reasoning using AWS fundamentals. It is a hypothetical architecture exercise, not a claim of a production deployment. The exercise focuses on selecting appropriate AWS services, applying basic security principles, and identifying operational considerations.

Scenario

A small web application needs a compute environment, object storage for files, controlled identity and permissions, network isolation, event-driven processing, and basic monitoring. The design should follow least-privilege access and keep sensitive resources appropriately restricted.

Proposed AWS Components

Requirement	AWS Service	Reasoning
Application compute	Amazon EC2	Provides configurable virtual servers for application workloads.
Object/file storage	Amazon S3	Provides durable object storage with access controls and lifecycle options.
Identity & permissions	AWS IAM	Controls users, roles, and permissions using least-privilege principles.
Network isolation	Amazon VPC	Provides logical network boundaries, subnets, and network security controls.
Event-driven tasks	AWS Lambda	Runs short event-driven functions without requiring server management.
Monitoring	Amazon CloudWatch	Collects metrics and logs and supports operational monitoring and alarms.

Security Considerations

1. Apply least-privilege IAM policies and prefer IAM roles for AWS resources rather than embedding long-lived credentials. 2. Restrict network access with VPC security controls and expose only required services. 3. Keep sensitive S3 data private and apply appropriate access controls. 4. Monitor relevant authentication, application, and infrastructure activity through logging and CloudWatch alarms. 5. Review permissions and resources periodically to reduce unnecessary access and cost.

Operational Checks

Before production use, I would verify network paths, IAM permissions, logging coverage, backup and recovery requirements, resource sizing, cost expectations, and failure scenarios. I would document assumptions and test the design before treating it as production-ready.

Skills Demonstrated

AWS fundamentals • Cloud architecture reasoning • IAM and security concepts • Networking concepts • Technical documentation • Structured problem-solving • Attention to detail

Portfolio note: This is an original technical exercise created to demonstrate reasoning and documentation. It does not contain confidential employer information and does not claim production ownership.